



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
In-Semester Examination II
Academic Year: 2024-25 (Semester – II)

Class: FY / M.Tech.	Subject: Advanced Concepts (PLM)	Subject Code: PCC-PL510
Date: 19.03.2025	Time: 11:00 to 11:45	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/sharing of stationery, calculator, etc. is not allowed.
- iv. Figures to the right indicate full marks.
- v. The symbols carry the usual meaning.

Q.N.		Marks	CO	BT
Q.1	a. What is workflow? Explain the following task a) Do Task b) Review Task c) Condition Task	07	CO3	BT1, BT2
	b. What are the different types of Lows? Explain it.	03	CO3	BT1, BT2
Q.2	a. Explain the 100%, 120% and 150% BoM. Explain the imprecise assembly.	07	CO3	BT2
	b. What is lifecycle visualization? Explain its capabilities.	03	CO4	BT1, BT2

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Exhibits the use of various elements of PLM.
CO 2	Selects proper architecture of PLM tools for an Industry.
CO 3	Administer for installing, configuring, integrating and efficient management of PLM tools.
CO 4	Select proper methodology for data transfer and implementation in CAD and PLM systems.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
ISE II Examination
Academic Year: 2024-25 (Semester – II)

Class: M.Tech (II Sem)	Subject: Internet of Things (Elective-IV)	Subject Code: PEC-PL-519
Date: 20.03.2025	Time: 11:00 to 11:45	Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/sharing of stationery, calculator, etc. is not allowed.
- Figures to the right indicate full marks.
- Indicate the Question Numbers while answering each question correctly.

Q.N.	Solve any TWO out of THREE sub-questions in Each .	Marks	CO	BT
Q.1 a	Write a Python program to calculate the hypotenuse of a right angled triangle.	5	CO3	BT3
b	Write an introduction to Raspberry Pi and explain its basic architecture with a neat labeled diagram. Explain the various components and stepwise procedure to flash the Operating System for Raspberry Pi. (03+02)	5	CO3	BT2
c	Draw and explain the SDN architecture. Also, compare SDN with traditional networking by highlighting 04 key differences. (03+02)	5	CO3	BT2
Q.2 a	What do you mean Data Handling and analytics? Give the classification of Data.	5	CO4	BT4
b	Explain briefly the concept of 'Cloud Computing.'	5	CO4	BT2
c	Illustrate and briefly explain the IoT-based Smart City application implemented on the cloud.	5	CO4	BT3

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

On successful completion of this course, students should be able to:

CO 3	To familiarize the programming languages such as Python and Raspberry Pi and extends its application to Software Defined Networking.
CO 4	To explore the relationship between IoT, Cloud Computing and Big Data.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
In-Semester Examination II
Academic Year: 2024-25 (Semester -II)

Class: M.Tech.

Subject: Machine Learning

Subject Code: PEC-PL-518

Date: 21.03.2025

Time: 11:00 to 11:45

Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. This paper contains one page.

Q.N.			Marks	CO	BT	
Q.1	For the following given data set (Only for Y _{test}), you have to calculate MAE, MSE, RMSE and R2 Score.		8	CO 1	BT 1-3	
		Actual				Predicted
	1	377				407
	2	122				126
	3	570				649
	4	632				630
	5	116				115
	6	109				107
Q.2	Describe Quartiles in detail and find out the Outliers for the following data set: 3, 14, 22, 19, 29, 70, 10, 49, 36, 32.		6	CO 1	BT 1-3	
Q.3	Normalize the dataset {45, 15, 75, 25, 55, 85, 35, 65} using Min-Max <u>Scaling</u> , Mean <u>Normalization</u> , and Max Absolute <u>Scaling</u> . Compute and write the <u>results</u> for each method.		6	CO 1-3	BT 1-3	

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Appreciate the underlying mathematical relationships within and across Machine Learning algorithms and the paradigms of supervised and un-supervised learning.
CO 2	Understand the concept behind neural networks for learning non-linear functions
CO 3	Design and implement various machine learning algorithms in a range of real-world applications.